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ES 399: PROJECTS IN ELECTRONICS I

TITLE: AUTOMATIC PHASSE CHANGE FROM 3 PHASE SUPPLY

A Project Report in Partial Fulfilment for the Award of Bachelor of Science in Electronic science
and communication

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ABSTRACT

This project presents the design and implementation of an Automatic Phase Changeover System from Three Phase Supply. The system is developed to ensure continuous power supply to a load by automatically selecting the available and healthy phase among the three phases (R, Y, B). In many power distribution systems, one or more phases may fail due to faults, voltage fluctuations, or outages, which can cause equipment damage and operational interruptions.

The proposed system continuously monitors the voltage level of each phase using a voltage sensing circuit and automatically switches the load to an active phase whenever the current phase fails or drops below a preset threshold. Relays and comparators (or a microcontroller-based control unit) are used to perform detection and switching operations.

This system improves reliability, protects electrical equipment from single phasing and under-voltage conditions, and reduces downtime in residential, commercial, and industrial applications.

DECLARATION

I, ARNOLD LUKAS KALIKENYE, with registration number 2023-04-03810, declare that the work described in this report is my original work. I wrote it with assistance and guidance from my supervisor, Dr. HUBERT MSUYA, with knowledge from different books, articles, and cite from the internet.

I declare that the work in this report was carried out in accordance with the regulations of the University of Dar es Salaam and has not been presented to any other University for examination.

SIGNED:

DATE:

This report may proceed for submission for assessment for the award of BSc. In Electronic Science and Communication at the University of Dar es Salaam.

Supervisor signature:

Date:

LIST OF ABBREVIATIONS

AC	Alternating Current
DC	Direct Current
R	Red Phase
Y	Yellow Phase
B	Blue Phase
V	Voltage
A	Ampere
Hz	Hertz
VSR	Voltage Sensing Relay
MCU	Microcontroller Unit
PCB	Printed Circuit Board
LED	Light Emitting Diode
SPDT	Single Pole Double Throw
DPDT	Double Pole Double Throw
NO	Normally Open
NC	Normally Closed
UPS	Uninterruptible Power Supply

CHAPTER ONE: INTRODUCTION

1.1 Introduction

Reliable electrical power supply is essential for the proper operation of residential, commercial, and industrial equipment. In many distribution systems, power is supplied in the form of a three-phase supply consisting of Red (R), Yellow (Y), and Blue (B) phases. Although three-phase systems are generally reliable, faults such as phase failure, voltage drop, or imbalance may occur. These problems can cause serious damage to electrical equipment, especially motors and sensitive electronic devices.

The Automatic Phase Change from Three Phase Supply system is designed to ensure continuous power delivery by automatically selecting an available and healthy phase whenever one phase fails or drops below the acceptable voltage level. Instead of manual switching, which may cause delay and inconvenience, the automatic system detects the condition of each phase using a voltage sensing circuit and transfers the load to the best available phase through relays or contactors.

This project focuses on designing and implementing a system that continuously monitors all three phases, compares their voltage levels with a preset reference, and performs automatic switching when necessary. The system improves power reliability, prevents equipment damage due to single phasing, reduces downtime, and enhances operational efficiency.

The automatic phase changeover system is suitable for homes, offices, hospitals, and industries where uninterrupted power supply is required.

1.2 Background of the Study

In the early days of electricity, power was supplied using single-phase systems. As industries grew, there was a need for more efficient and reliable power systems. This led to the introduction of the three-phase system, which is more efficient for transmitting and using electrical power, especially in running motors and heavy machines.

The three-phase system was developed in the late 19th century and later became widely used in industries, homes, and commercial buildings because it provides stable and continuous power. However, over time, problems such as phase failure, voltage drop, and power imbalance were observed. These problems caused damage to electrical equipment and interrupted operations.

To solve these issues, engineers first used manual phase changeover switches. But manual switching was slow and inconvenient. Later, automatic systems were developed using relays and electronic circuits to detect faulty phases and switch to a healthy phase automatically.

Today, automatic phase changeover systems are commonly used to ensure continuous power supply and protect electrical devices from damage. This project is based on this development and focuses on designing a simple and reliable automatic phase change system.

1.3 Problem Statement

In many residential, commercial, and industrial areas, electrical power is supplied through a three-phase system (R, Y, B). Although this system is reliable, one or more phases may fail due to faults, maintenance issues, overload, or voltage fluctuations. When a phase fails or its voltage drops below the required level, electrical equipment may stop working or become damaged.

In most cases, phase changeover is done manually. Manual switching causes delays, requires human intervention, and may lead to equipment damage due to single phasing or under-voltage conditions. This is especially dangerous for motors, sensitive electronics, and continuous production systems.

Therefore, there is a need for an automatic system that can continuously monitor all three phases and automatically switch the load to a healthy phase whenever a failure occurs. The system should operate quickly, safely, and reliably to ensure continuous power supply and protect electrical equipment from damage.

1.4 Objectives of the Report

1.4.1 Main Objective

The main objective of this project is to design and implement an Automatic Phase Changeover System from a Three-Phase Supply that can automatically detect phase failure or voltage drop and switch the load to an available healthy phase without manual intervention, ensuring continuous power supply and protection of electrical equipment.

1.4.2 Specific objectives

- i. To design and implement a reliable three-phase voltage monitoring system for real-time detection of phase conditions.
- ii. To develop an automatic control circuit that detects phase failure and under-voltage faults.
- iii. To construct an efficient automatic switching system that selects a healthy phase instantly and safely.
- iv. To enhance continuous power supply and protect electrical equipment from damage caused by phase instability.

1.5 Significance of the report

- I. To provide a practical solution to power supply interruptions caused by phase failure and voltage instability in three-phase systems.
- II. To improve reliability by ensuring automatic switching to a healthy phase without human intervention.
- III. To protect electrical equipment such as motors and sensitive electronic devices from damage caused by single phasing and under-voltage conditions.
- IV. To reduce downtime in residential, commercial, and industrial applications, thereby improving efficiency and productivity.

V. To enhance understanding of automatic control systems and the application of electronic components in power management.

VI. To offer a cost-effective design that can be implemented in real-life power distribution systems.

1.5 Scope and Limitations

1.5.1 Scope

I. The project focuses on designing an automatic phase changeover system for a three-phase power supply.

II. The system is designed to monitor voltage levels of the three phases and automatically switch to a healthy phase when failure or under-voltage is detected.

III. The project is mainly implemented using basic electronic components such as voltage sensing circuits, comparators, and relays.

IV. The system is suitable for protecting small and medium electrical loads such as domestic and light industrial equipment.

1.5.2 Limitations

I. The system does not provide protection against overcurrent, short circuit faults, or phase sequence reversal.

II. It is designed for low to medium power applications and may require modification for high-power industrial loads.

III. The system may have slight switching delay depending on component response time.

IV. The design does not include remote monitoring or advanced microcontroller-based control features unless upgraded.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Automatic phase changeover systems have been developed to improve reliability in electrical power distribution. Different researchers and engineers have studied methods of detecting phase failure and automatically switching loads to a healthy phase.

The three-phase power system was introduced following the work of Nikola Tesla, who contributed significantly to the development of alternating current (AC) power systems. Three-phase supply is widely used because it provides continuous power delivery and is more efficient for running motors compared to single-phase supply.

According to electrical protection studies, phase failure is one of the major causes of motor damage in industrial applications. When one phase fails, the motor continues to operate on two phases, leading to overheating and possible winding burnout. Automatic protection systems are therefore necessary to prevent such problems.

Previous designs of phase changeover systems mainly used electromechanical switches and manual transfer switches. These systems required human intervention and were not suitable for fast response during power failure.

Modern research has introduced electronic phase monitoring circuits using components such as comparators, sensors, and relays. For example, the use of operational amplifiers like LM358 allows voltage comparison between phase signals and reference voltage levels. When the sensed voltage drops below the reference level, the control circuit activates the relay to switch the load to another healthy phase.

Furthermore, automatic transfer systems are widely used in power backup systems to ensure uninterrupted supply. Studies show that automatic switching improves system reliability, reduces downtime, and protects electrical equipment from damage caused by single phasing and voltage instability.

In conclusion, literature shows that automatic phase changeover systems are essential in modern power management, and the use of electronic control circuits provides an efficient and reliable solution.

2.5 Relates works

Several methods have been developed to improve reliability in three-phase power systems.

Early systems used manual changeover switches, where an operator manually selected a healthy phase when one failed. However, this method was slow and unsafe.

Later, relay-based automatic phase changeover systems were introduced. These systems used voltage-sensing relays to detect phase failure and automatically switch to another available phase. Although automatic, they suffered from mechanical wear and slower response.

To improve reliability, electronic (static) phase monitoring circuits were developed using comparators and semiconductor devices. These systems were faster and more durable than mechanical relays.

Modern systems use microcontrollers (such as Arduino or PIC). These monitor phase voltages digitally, apply programmed logic, and control relays. They provide better accuracy, flexibility, and additional protection features.

In industrial applications, PLC-based systems are used for advanced monitoring, fault detection, and integration with control systems.

This project is based on automatic phase detection and switching using electronic sensing and relay control to provide reliable and continuous power supply at low cost

2.6 Project gaps and research

From the related works reviewed:

- Manual systems require human intervention.
- Relay-based systems suffer from slow response and mechanical wear.
- Some electronic systems lack proper phase priority control.
- Many systems do not include under-voltage protection.
- PLC and smart systems are expensive and complex for small-scale users.
- Some designs do not protect against phase conflict (two phases connected at once).

Identified Gap

There is a need for a:

- Low-cost
- Reliable
- Fast-switching
- Safe automatic phase changeover system
- With proper phase priority logic
- With under-voltage detection
- Suitable for residential and small industrial use

How This Project Fills the Gap

This project:

- Automatically detects phase failure
- Uses voltage sensing and comparator circuits
- Provides phase priority selection ($R \rightarrow Y \rightarrow B$)
- Prevents phase conflict
- Disconnects load if all phases fail

Is simple, affordable, and reliable

2.7 Linkage to the Present Study

Previous studies have shown that automatic phase changeover systems improve power reliability by detecting phase failure and switching to an available phase. Manual systems were inefficient and unsafe, while relay-based and electronic systems improved automation but had limitations such as mechanical wear, limited flexibility, and high cost in advanced PLC-based systems.

The present study builds upon these earlier works by designing a simple, low-cost, and reliable automatic phase changeover system using voltage sensing, comparator circuits, and relay control.

Unlike manual systems, this study ensures fully automatic operation. Compared to complex PLC-based systems, it provides an affordable solution suitable for residential and small industrial applications.

Additionally, this study introduces:

- Proper phase priority selection ($R \rightarrow Y \rightarrow B$)
- Under-voltage detection
- Load protection during total phase failure
- Safe switching to prevent phase conflict

CHAPTER THREE: METHODOLOGY

3.1 Introduction

In three-phase electrical systems, phase failure can lead to equipment damage or inefficient operation. This project aims to design and implement a system capable of detecting phase failure and automatically switching to an alternative phase, ensuring continuous operation of the connected load. The design process followed a systematic approach involving analysis, circuit design, simulation, construction, and validation.

3.2. System Design and Development

3.2.1 System Analysis

The first step involved analyzing the behavior of the three-phase power system. Key objectives included:

- Studying the dynamics of three-phase power systems.
- Identifying common causes of phase failure, such as open circuits or imbalanced loads.
- Determining the switching priority, which was established as: Red (R) → Yellow (Y) → Blue (B).

3.2.2 Block Diagram Design

The system's architecture was divided into the following blocks:

1. **Voltage Sensing Unit:** To monitor the health of each phase.
2. **Comparator Unit:** To compare phase voltages with a reference voltage.
3. **Relay Driver Circuit:** To control relays based on comparator outputs.
4. **Switching Unit (Relays/Contactors):** To switch between phases.
5. **Load:** The connected load that is powered by the system.

3.2.3 Hardware Design

The hardware design was developed in several stages, focusing on the creation of critical circuits required for phase detection and switching.

i. Voltage Sensing Circuit

Each phase voltage was stepped down using a **220V/12V transformer**. The AC voltage was then rectified using a **bridge rectifier** to convert the alternating current into direct current (DC). A **filter capacitor** was employed to smooth the DC voltage, which was used as an indicator of phase health.

ii. Comparator Circuit

The **LM358 operational amplifier** was utilized as a comparator. The voltage of each phase was compared against a reference voltage (V_{ref}). The decision logic for comparison is as follows:

If $V_{phase} > V_{ref}$, the phase is deemed healthy.

If $V_{phase} < V_{ref}$, the phase is considered faulty.

iii. Relay Driver Circuit

A **transistor** was used to amplify the current from the comparator's output, driving the relay coil. Additionally, a **flyback diode** was placed across the relay coil to prevent voltage spikes, ensuring the protection of sensitive components.

iv. Switching Mechanism

The switching mechanism involved using **power relays/contactors** to connect the load. The design ensured that only one phase could be connected at a time. **Electrical interlocking** mechanisms were implemented to prevent two phases from being connected simultaneously, safeguarding the system from potential short circuits.

3.3 Control Logic Implementation

The system's control logic was designed with a priority sequence for phase switching:

- If Phase R is healthy, it is connected to the load.
- If Phase R fails, the system switches to Phase Y.
- If Phase Y fails, the system switches to Phase B.
- If all phases fail, the system disconnects the load to prevent further damage.

3.4 Simulation

The system was simulated using **Proteus** software to verify the following:

- Correct detection of phase health.
- Accurate switching logic based on phase priority.
- Prevention of short circuits between phases during switching.

3.5 Testing and Validation

The system will be tested by:

- Manually disconnecting one phase to simulate phase failure.
- Observing the automatic switching behavior of the system.
- Measuring the output voltage at the load to confirm proper operation.
- Ensuring no simultaneous connection of multiple phases, adhering to safety protocols.

CHAPTER FOUR: SYSTEM DESIGN AND ANALYSIS

4.1 Introduction

The system is designed in the following stages:

I. Input Stage (Three-Phase Supply)

Input: R, Y, B phases + Neutral

Each phase supplies about 220–240V AC (line to neutral).

The system selects only one phase at a time to feed the load.

II. Voltage Sensing Design

Each phase is connected to:

Step-down transformer (e.g., 220V/12V)

Bridge rectifier (AC to DC conversion)

Filter capacitor (smooth DC)

Voltage divider (reduce voltage level)

- Purpose: To convert high AC voltage into low DC voltage suitable for comparison.

III. Comparator Design

Component used:

LM358

Design principle:

One input = sensed DC voltage

Other input = reference voltage

If:

$V_{\text{sense}} > V_{\text{ref}} \rightarrow \text{Output HIGH}$

$V_{\text{sense}} < V_{\text{ref}} \rightarrow \text{Output LOW}$

This determines whether the phase is healthy or failed.

IV. Relay Driver Design

Since comparator output current is small:

A transistor (e.g., NPN) is used

Base resistor limits current

Diode is placed across relay coil (flyback protection)

Relay coil voltage: 12V DC

When transistor turns ON $\rightarrow$ Relay energizes $\rightarrow$ Phase connects to load.

V. Phase Priority Logic

Priority arrangement example:

R phase Highest priority

Y phase Second

B phase

Third

If R fails system automatically activates Y

If Y fails system switches to B

- System Analysis

A. Voltage Analysis

Example:

Transformer output = 12V AC

After rectification:

After filtering $\approx 15\text{--}16\text{V DC}$

Voltage divider reduces to $\approx 5\text{V}$ for comparator input.

B. Power Analysis

Relay coil power:

If relay = 12V, 50mA

System power consumption is low.

C. Switching Time Analysis

Relay switching time $\approx 10\text{--}20\text{ ms}$

This ensures fast changeover without major interruption.

D. Reliability Analysis

-Automatic operation

-No manual intervention

-Protects equipment from low voltage

-Prevents two phases from connecting simultaneously

- Overall Performance

The designed system:

Detects phase failure accurately

Switches automatically

Maintains continuous supply

Operates safely and reliably

If you want, I can now give you:

- Circuit diagram explanation step by step
- Limitations of the system
- Oral defense explanation version
- Complete Chapter Three (System

4.3 System components

I. Three-Phase Supply Source

R, Y, and B phases with neutral. It provides the main AC input power to the system.

II. Step-Down Transformers (Three Units)

Used to reduce high AC voltage (220–240V) to a lower AC voltage (e.g., 12V) suitable for sensing and control circuits.

III. Bridge Rectifiers

Convert the stepped-down AC voltage into DC voltage.

IV. Filter Capacitors

Smooth the rectified DC voltage by reducing ripple.

V. Voltage Divider Resistors

Reduce the DC voltage to a safe level for the comparator input.

VI. Comparator Integrated Circuit

LM358

Compares the sensed voltage with a reference voltage and determines whether a phase is healthy or faulty.

VII. Reference Voltage Source

Provides a constant voltage used for comparison in the comparator circuit.

VIII. Transistors (Relay Drivers)

Amplify the comparator output current to drive the relay coils.

IX. Relays (Three Units)

Used for switching the load to the selected healthy phase. They ensure only one phase is connected at a time.

X. Flyback Diodes

Protect transistors from voltage spikes generated by relay coils.

XI. Indicator LEDs

Show the active phase and indicate phase failure.

XII. Load

Single-phase appliance such as a bulb, motor, or other electrical equipment.

CHAPTER FIVE: SYSTEM IMPLIMENTATION

5.1 Introduction

The system implementation describes how the designed circuit is physically constructed, connected, and tested to achieve automatic phase switching.

I. Hardware Construction

The components were assembled on a circuit board. Each phase (R, Y, B) was connected to its own sensing circuit consisting of a step-down transformer, bridge rectifier, filter capacitor, and voltage divider.

The control section using the comparator IC LM358 was connected after the sensing stage.

II. Voltage Sensing Implementation

Each phase voltage was reduced from 220–240V AC to 12V AC using a transformer.

The AC voltage was then converted to DC using a bridge rectifier and filtered using a capacitor.

The output DC voltage was fed into the comparator input for monitoring.

III. Comparator and Control Section

The LM358 comparator compared each sensed voltage with a preset reference voltage.

If the sensed voltage was above the reference level, the comparator output became HIGH, indicating a healthy phase.

If below the reference level, the output became LOW, indicating phase failure.

IV. Relay Driver Implementation

The comparator output was connected to an NPN transistor through a base resistor.

When the comparator output was HIGH:

The transistor turned ON

The relay coil was energized

The corresponding phase was connected to the load

A flyback diode was placed across each relay coil for protection.

V. Phase Priority Arrangement

The relays were wired in such a way that:

R phase had first priority

Y phase had second priority

B phase had third priority

If R failed, the system automatically switched to Y.

If Y also failed, the system switched to B.

VI. Load Connection

The output of the relay contacts was connected to a single-phase load (such as a bulb).

Only one relay could be active at a time to prevent short circuits between phases.

VII. System Testing

The system was tested by:

Disconnecting one phase manually

Reducing voltage below threshold

Observing automatic switching

The results showed successful automatic phase change with minimal delay.

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